# Exercises Analytics and Knowledge Discovery WS 2025

September 8, 2025

FH Salzburg
Informationstechnik &
System-Management

## Data preparation

DP1 **Handling missing data by removal of data records**:
Consider the dataset dataIsRemovalAllowed which has the following three columns: gender (binary coded), age and preference score for a product. The goal is to assess the mean preference for a product (the preference is a score between 0 and 10).

(a) Can rows with missing data be removed without significantly influencing the result?
The answer is NO. Explain graphically, why nonrandom missingness of data here would lead to an underestimation of the preference score of the product.

DP2 **Handling missing timeseries data using data imputation methods**
Load the dataset ex22PM10imputation containing timeseries of PM10 values in Lehen of one year. The first and the second column contain the same timeseries, but in the first column, 300 values are missing (entries Nan where Nan stands for Not a number). The goal is to impute 300 values using different methods and compare them to the real ones (in the second column).
Program and apply the following 4 methods for data imputation.
Hint1: the time interval is 30 minutes, so a day consists of 48 values.
Hint2: to make programming easier, values are only missing one at a time, i.e. there are no missing values following each other.

(a) Substitute with the simple but often bad method that **takes the mean value of the whole time series** (generic method not only applicable for timeseries).

(b) Substitute the Nan values by the simplest **extrapolation** method, i.e., by taking the **last known value**.

(c) Substitute by **linear interpolation** using the last known and the next known value.

(d) Substitute with a **similar datetime approach** by taking the **value of the day before** at the same time.

DP3 **Usage of EDA for discussion of the previous data imputation results**
Describe the results of the data imputation example before using EDA. Answer the following questions using appropriate graphics.
An important, recurring topic in data analysis is evaluation, especially computation and description of errors: besides choosing suitable graphics (suitable for the question to be answered) one also needs to choose a suitable error measure and often also a reference for sake of comparison.

(a) Which of the 4 tried data imputation methods is best suited for this timeseries? Is the winner the expected one? For the following 3 questions choose this best method.

(b) Is the imputation error high? Can it be expected that imputation does not harm the following analysis?

(c) What is the **distribution** of the imputation error? What is the probability, that the relative error is more than for example 5%? (Hint: you will probably need 2 graphics here)

(d) Plot the whole timeseries and add the imputed data points. This plot is often used to discuss if imputation is expected to harm following analyses or not.

DP4 **Handling missing data by data imputation with extrapolation using regression based on other variables**

(a) Now we try another method: so first choose variable(s) that have a high correlation with the PM10 values in Lehen.

(b) Then apply data imputation to the missing PM10 values using a linear regression with these other variables.

(c) Compare the result of this imputation method with the imputed values from the last-value-method by making a suitable graphic.

DP5 **Feature selection based on high correlation** in Python
Look at `https://machinelearningmastery.com/feature-selection-for-regression-data/`
Redo the first part of the analysis done there:

(a) Run the code using features selected based on correlation of the feature with the outcome and explain what is done there.

(b) Run the code using features selected based on the mutual information of the feature and the outcome and explain what is done there.

(c) The important decision here is to **select the number of features k**. Do this **manually** for both methods by running the code with k= 5, 10, 20, 30, 40, 50, 60, 70, 80, 90. Then draw the resulting errors depending on k.
Which of the two methods seems to be better? Which k should we select?

DP6 **Feature generation to make linear regression models more powerful**
Load the data dataFeatureGenerationForLinReg.csv. The first two columns are the input variables $x_1$ and $x_2$, the third is the outcome $y$ which should be predicted from $x_1$ and $x_2$.

(a) Predict the outcome $y$ from $x_1$ and $x_2$ using a purely linear model and assess the fit graphically.

(b) Predict the outcome $y$ from $x_1$ and $x_2$ but now add polynomial terms to the linear model. Again assess the fit graphically. Is this model better than the one before?